A PROJECT TITLE

ON

Loan Approval Prediction Using Ensemble Learning (RF, XGBoost , LightGBM)

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

Name : S.Koushik roll : 2520090113

Name : Ch.Ganesh roll : 2520090112

Under the Esteemed guidance of

Dr.Y. Lakshmi Prasanna

Asst.Professor , CS&IT

CS&IT



KLH (DEEMED TO BE UNIVERSITY)

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CS&IT

Name of Title: Loan Approval Prediction Using Ensemble Learning (RF, XGBoost, LightGBM)

1. Abstract:

The increasing demand for quick and accurate loan approval decisions has encouraged financial institutions to adopt intelligent machine learning techniques for risk assessment. Traditional loan approval methods are often time-consuming, prone to human error, and may fail to identify complex patterns in applicant data. This project proposes an automated Loan Approval Prediction System using alearning approach that combines three powerful machine learning algorithms: Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LightGBM). The primary objective is to predict whether a loan application should be approved or rejected based on various applicant attributes such as age, income, employment status, education, marital status, credit history, loan amount, loan term, and property area.

The proposed system begins with data preprocessing, including handling missing values, encoding categorical variables, feature scaling, and exploratory data analysis. The dataset is then divided into training and testing sets for model development and evaluation. Each algorithm is trained individually, and their performance is compared using evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC Score. Hyperparameter tuning is performed to improve prediction accuracy and reduce overfitting.

Among the evaluated models, the algorithm with the best performance is selected for deployment in the prediction system. The proposed approach aims to provide reliable and unbiased loan approval decisions while minimizing financial risk for lending institutions. By leveraging ensemble learning techniques, the system achieves higher prediction accuracy and better generalization compared to traditional machine learning methods. This project demonstrates how advanced machine learning models can support banks and financial organizations in making faster, data-driven, and more consistent lending decisions, ultimately improving operational efficiency, reducing default risk, and enhancing customer satisfaction.

2. INTRODUCTION:

Loan approval is a crucial process in the banking sector that affects customer satisfaction and financial stability. Traditional manual approval methods are often time-consuming, error-prone, and influenced by subjective decisions. To address these challenges, this project proposes a machine learning-based loan approval prediction system using ensemble learning techniques.

The system employs Random Forest (RF), XGBoost, and LightGBM to predict loan approval based on applicant details such as income, employment status, credit history, and loan amount. Data preprocessing includes handling missing values, encoding categorical features, feature selection, and normalization. The models are evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC, and the best-performing model is selected for prediction.

This approach improves prediction accuracy, reduces processing time, minimizes human bias, and supports consistent, reliable loan approval decisions, demonstrating the effectiveness of ensemble learning in financial applications.

3. SOFTWARE REQUIREMENTS:

Programming Language: Python 3.10+

Development Environment: Colab / Visual Studio Code

Machine Learning Libraries: Scikit-learn, XGBoost, LightGBM

Data Processing Libraries: NumPy, Pandas

Visualization Libraries: Matplotlib, Seaborn

Signature of the Guide

Course Instructor